The Polarization Penalty: Partisan Intense Language and Online Engagement

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I. Introduction

Social media platforms promote content based on engagement. If polarizing language receives high engagement, this may create echo chambers where polarizing ideas are boosted (Barberá, 2020). Additionally, this type of content may spread misinformation more quickly or distort public perception of current events. If users know this type of content receives more engagement, it may incentivize them to post more polarizing content.

During COVID-19 lock-downs, social media sites like Twitter (X), Facebook, and Reddit became the only social space for many individuals. These platforms were used for social interaction but also for news, with people particularly engaged with political issues related to the pandemic such as masking, vaccines, and lock-downs. The World Health Organization (WHO) has since labeled this period as an "Infodemic," describing the overload of information available online - both accurate and inaccurate (World Health Organization, 2020).

Existing literature shows social media has become a significant channel for diffusion of political content and engagement-driven platforms can drive the dispersion of polarizing content (Allcott and Gentzkow, 2017). Although platforms facilitate content diffusion, Gentzkow and Shapiro (2011) indicate user demand drives exposure and users generally prefer a broad range of content as opposed to an echo chamber. Contrastly, Vousoughi et al. (2018) observe a novelty premium for false and polarizing content that drives dispersion.

This study investigates if there was an engagement reward or penalty for political polarization on Reddit posts during the COVID-19 pandemic. Namely, I implement a Double Machine Learning (DML) and Causal Forest framework to identify the causal effect of polarization on Reddit post engagement. This study contributes to the literature by quantifying the causal impact of polarization on engagement, considering heterogeneity in this relationship, and examining a period characterized by high levels of online political engagement.

My results indicate that Reddit content posted during the COVID-19 pandemic faced a polarization penalty on average, where posts were less engaged with if they had more polarization. Specifically, a 1% increase in polarization corresponds to a 0.13% decrease in engagement.

However, this penalty is not uniformly distributed. Posts with highly urgent rhetoric have a larger penalty of -0.19% whereas low urgency posts have a polarization reward of 0.02%. Moreover, if a post is generally neutral, the polarization penalty is only -0.06% but nearly doubles to -0.11% for posts with high political salience. There is also an asymmetric element where left-wing posts have a higher penalty of -0.20% compared with right-wing posts that have a -0.17% penalty.

These results suggest that in a period with high social media use and where a significant portion of daily life was politicized, user demand for reliable information outweighed any novelty premium of polarizing or extreme content. This finding provides a contrasting view to the common assumption that social media drives the spread of extreme or polarizing content. Moreover, it supports the literature that user demand can act as a moderating force, acting as a tax for polarizing content.

II. Data

I use the Reddit COVID dataset sourced from Kaggle (pavellexyr, 2021), which contains 4,510,178 Reddit posts related to COVID-19 from the pandemic's onset up until 25 October 2021. Posts must mention COVID-19 in the title to have been included in the sample. For each post in the sample, the dataset contains its title, body, subreddit posted in, time and date of posting, and the sum of the posts up and down votes presented as an overall score.

The overall text for a post is the combination of the title of the post and the body of the post, if available. Where posts have been deleted or removed, the dataset still contains the title of said posts. However, given that a post's engagement is frozen following deletion or removal, I exclude these posts from the sample.

For computational feasibility, I randomly sample 100,000 posts. In this subsample, there are 11,754 unique subreddits. Further, I reduce the sample to only include the top 100 subreddits, which improves the reliability of subreddit fixed effects and limits sparsity from subreddits with low-activity. The resulting sample is 55,821 posts.

To determine the political affiliation of a post, I classify the text using valhalla/distilbart-mnli-12-3 model from the Hugging Face Transformers library (Lewis et al., 2020). The model produces a probability score representing the best estimate of if the text belongs in the right-wing, left-wing, or neutral category.

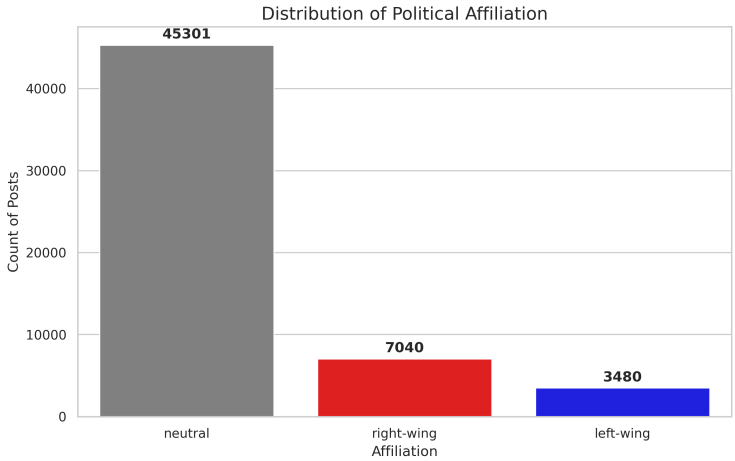


Figure 1: Political Affiliation of Posts in Sample

Using these scores, I compute the political salience of a post (PS_i). This measure captures the amount of political content in a post, regardless of political direction:

$$PS_i = 1 - \hat{P}(Neutral_i|Text_i) \times 100.$$

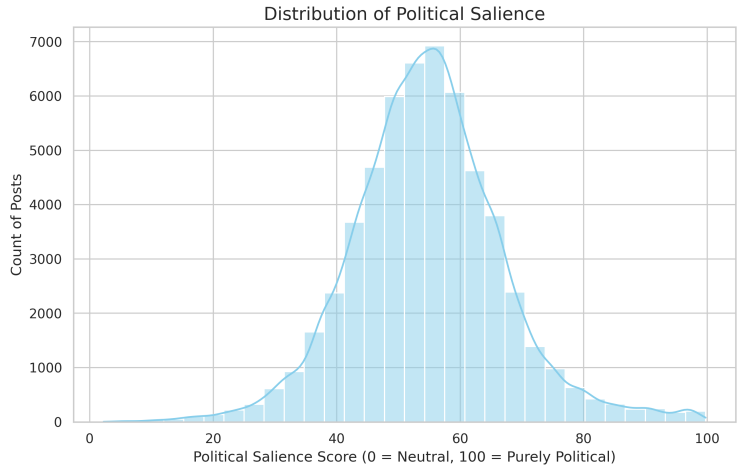


Figure 2: Political Salience

Next, using a normalization approach similar to Gentzkow and Shapiro (2010), I compute the political polarization (PP_i) of a post. This

measures the relative balance between right- and left- leaning rhetoric:

$$PP_i = \frac{P(RightWing_i|Text_i) - P(LeftWing_i|Text_i)}{P(RightWing_i|Text_i) + P(LeftWing_i|Text_i)} \times 100.$$

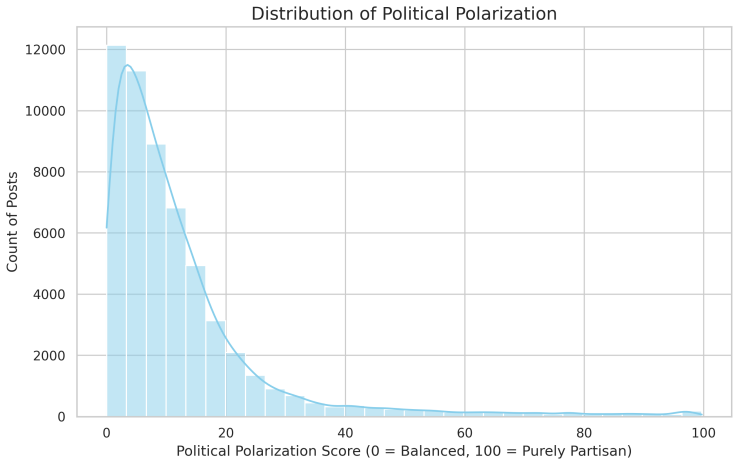


Figure 3: Political Polarization

Engagement of a post is measured as the the sum of up-votes and down-votes on a post. I have not considered comments on a post since they are sensitive to moderator intervention and individuals are not limited to one comment, which is the case for up-vote versus down-vote. The engagement scores show a heavy right-tail, which is common in social media datasets. As can be seen in Figure 4, I perform a log-transformation which successfully compresses the variance for usability. However, the distribution maintains a spike for low engagement levels, reflecting the nature of social media where the majority of posts receive little to no engagement.

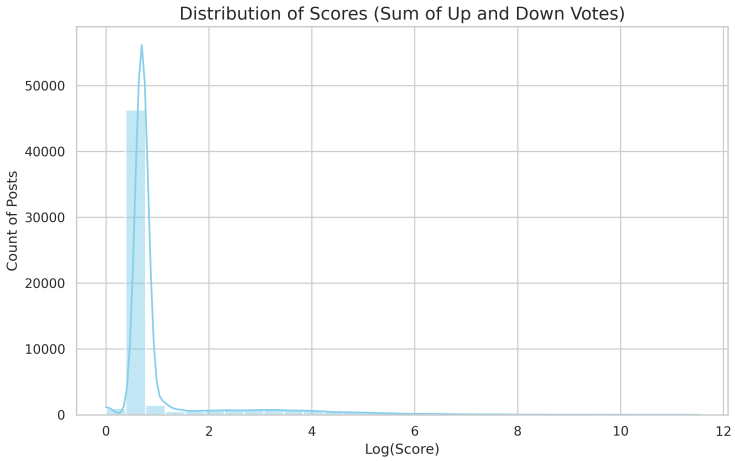


Figure 4: Engagement

Additionally, I create variables for the length of a post, urgency, and sincerity. Urgency is measured as the sum of the count of urgent words[1], exclamation marks, question marks, and capital letters in a post. Sincerity is measured as the number of sincere words[2] in a post.

Table 1: Descriptive Statistics for Continuous Variables

	Count	Mean	Std	Min	Max
Post Length	55,821	134.11	416.95	6.00	29,895
Urgency	55,821	10.04	17.60	0.00	1,277
Sincerity	55,821	0.13	0.88	0.00	58.00
Political Salience	55,821	54.71	12.18	2.24	99.74
Partisan Score	55,821	12.28	14.43	0.00	99.70
Engagement	55,821	54.64	1,204.91	0.00	99,881
Log(Engagement)	55,821	1.05	1.12	0.00	11.51

III. Empirical Approach

To determine the causal effect of polarization on user engagement, I use a Double Machine Learning (DML) and a Causal Forest framework. The DML approach removes the effect of control variables from both engagement and polarization before estimating the relationship between the two. As a result, the model isolates the variation in engagement that is attributable to polarization only. Further, the Causal

[1] breaking, urgent, now, alert, warning, attention, critical, must, stop

[2] we, me, I, ours, us, my, mine, our

Forest estimates specific treatment effects to identify heterogeneous effects within the sample. Both models use a configuration of 300 trees, maximum depth of 12, and minimum leaf size of 5 determined through randomized search during DML tuning.

Double Machine Learning (DML)

Following Chernozhukov et al. (2018), I use the partially linear regression model:

$$Y_i = \beta PP_i + g(X_i) + \epsilon_i$$

$$PP_i = m(X_i) + \mu_i$$

where Y_i is the log of a post's engagement, PP_i is the continuous treatment, polarization, and X_i gives controls including post length, sincerity, urgency, time of posting, political salience, and political affiliation. Additionally, I control for the subreddit location of each post. I do not explicitly control for topic variation or user-level characteristics which could influence polarization and engagement. However, the inclusion of subreddit fixed effects and text-based controls helps to mitigate some of the potential omitted variable bias.

The DML framework implements in two stages. First, the model estimates nuisance parameters using random forests and obtains the conditional expectation of Y and PP given X. Hence, the random forests separately predicts engagement and the polarization using the specified controls. Next, the residuals from this estimation are calculated for engagement and partisan intensity:

$$\hat{Y_i} = Y_i - \hat{g}(X_i)$$

$$\hat{PP_i} = PP_i - \hat{m}(X_i).$$

From the residuals, $\hat{Y_i}$ is regressed on $\hat{PP_i}$, isolating the variation in engagement that is only attributable to polarization and obtaining the causal effect, β. This coefficient is interpreted as the percentage change in engagement associated with a 1% increase in partisan intensity.

This approach is required due to the large number of controls, most notably the 100 subreddit fixed effects. In a traditional ordinary least squares (OLS) model, this high-dimensional nature can lead to overfitting or bias from misspecification of the functional form.

Causal Forest for Heterogeneity

As the DML model only provides the average treatment effect (ATE), I also apply a Causal Forest (Wager and Athey, 2018) to determine conditional average treatment effects (CATE). Hence, I can identify if the effect of polarization differs across categories such as political affiliation or urgency.

The Causal Forest extends a random forest estimator by dividing the data into trees that maximize the difference in the treatment effect across branches. The approach is non-parametric and ensures that any heterogeneity identified in the impact on engagement from partisanship is a result of the data structure and statistical robustness. For each condition or category, the observations are split in half. Using the first half, the model decides where to split the data to observe any differing effect. The second half is used to calculate the actual treatment effect for the category.

In a standard OLS framework with heterogeneity analysis, if all variables are interacted with the treatment, the model loses degrees of freedom and the standard errors are large. However, there may be bias in researchers selecting only certain variables to determine heterogeneous effects. The Causal Forest approach searches through all possible interactions and identifies the important conditions. Additionally, with the Causal Forest, the relationship does not have to be linear, as in OLS.

IV. Results

Table 1 presents results for the ATE calculated by the DML model. As can be seen, the effect is negative where a 1% increase in partisan intensity leads to a 0.13% drop in engagement. Hence, there is a statistically significant polarization penalty.

Table 2: Average Treatment Effect (ATE)

	Coefficient	Std. Err.	t-stat	$P > \|t\|$	2.5 %	97.5 %
Partisan Intensity	-0.0013	0.0005	-2.5654	0.0103	-0.0022	-0.0003

Table 2 shows the DML results compared with an OLS model. Both models produce a negative coefficient for partisan intensity but the DML's coefficient is over four times as large as the OLS model's coefficient. Additionally, the OLS model's result is not statistically different from zero. Hence, the OLS model likely suffers from selection bias or non-linear confounding. Since Reddit is a complex social interface, the variables involved likely have dynamic interaction and thus, require better prediction of the confounders and their relationship with high engagement. To reveal the observed polarization penalty, a more flexible machine learning approach is required.

Table 3: OLS versus DML Model Results

Model	Coefficient	Std. Err.	t-stat
OLS	-0.0003	0.0003	-0.9627
DML	-0.0013	0.0005	-2.5654

Table 3 explores heterogeneity where I aggregate treatment effects from the Causal Forest across key subgroups. For all categories, with the exception of low urgency, the effect of partisan intensity is negative. For low urgency, there is a slight 0.02% positive effect, which is statistically significant but very small. In contrast, high urgency posts have one of the steepest penalties compared with other subgroups. This result could indicate that partisan language facilitates discussion when using calmer words while with urgent words, partisan language creates toxic spaces that users avoid.

For sincerity, the split is defined at 0.0 implying the largest difference in treatment effects is posts with zero sincere words and posts with some sincere words. Posts with no sincere words have a marginally lower partisan penalty implying that users maybe slightly prefer objective rhetoric.

Posts with higher political salience lose more engagement for partisan language than posts with more neutral content. At low levels of political salience, the partisanship penalty is small at -0.06%. As the posts become more political, the penalty almost doubles to -0.11%. For left-wing posts, there is a harsher penalty of a 0.20% drop in engagement for a 1% increase in partisan intensity than for right-wing posts where the drop is only 0.17%. Audiences of left-leaning COVID-19 content may expect more neutral posting whereas right-leaning audiences have more tolerance for partisan messaging.

Table 4: Conditional Average Treatment Effect (CATE) by Subgroup

Feature	Group	Mean	Std Dev	Observations	t-stat
Left-wing	No	-0.0008	0.0106	52341	-17.3819
Left-wing	Yes	-0.0020	0.0057	3480	-20.4010
Right-wing	No	-0.0008	0.0108	48781	-15.5501
Right-wing	Yes	-0.0017	0.0063	7040	-22.3242
Urgency	Low (≤ 7.0)	0.0002	0.0120	28072	2.3059
Urgency	High (> 7.0)	-0.0019	0.0083	27749	-38.9162
Political Salience	Low (≤ 54.5)	-0.0006	0.0112	27917	-9.4044
Political Salience	High (> 54.5)	-0.0011	0.0095	27904	-19.8487
Sincerity	Low (≤ 0.0)	-0.0009	0.0105	52034	-18.8354
Sincerity	High (> 0.0)	-0.0010	0.0086	3787	-7.4850

Figure 5 reveals that urgency and political salience are the primary drivers of treatment heterogeneity. Hence, polarizing discourse may

be moderated by the nature of the tone used and topical relevance as opposed to the political leaning. Figure 6 provides the distribution of the CATE, showing the substantial spread. It suggests that the average user may be unresponsive but some specific segments exhibit higher sensitivity to polarizing sentiment.

Figure 5: Feature Importance Determined by Causal Forest

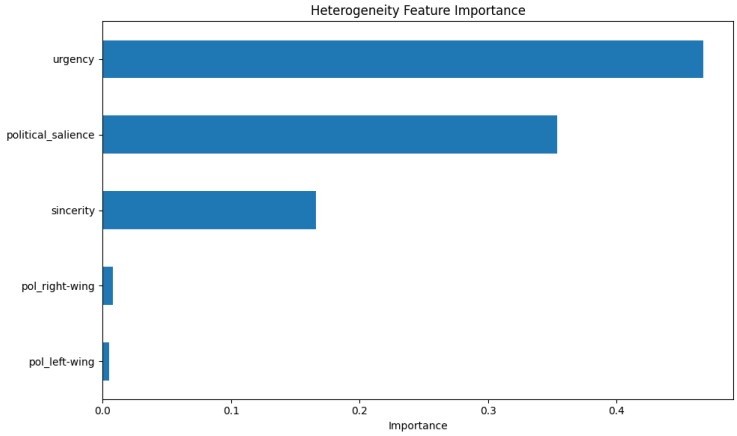
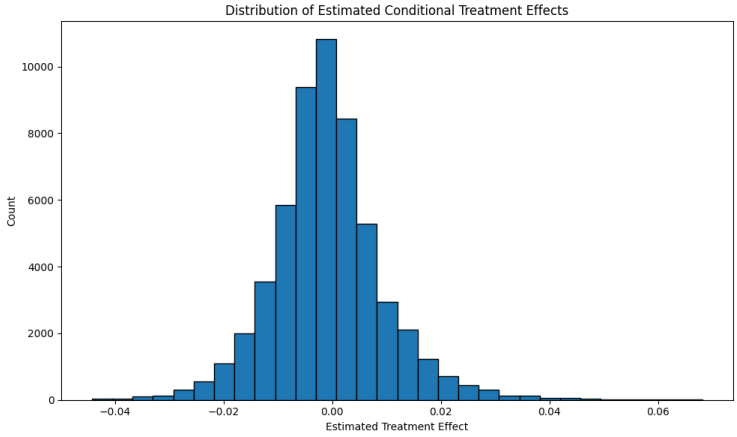


Figure 6: Treatment Effect Distribution



V. Conclusion

This paper uses Double Machine Learning (DML) and Causal Forest methods to show that during COVID-19, in mainstream subreddits, partisan language decreased user engagement with posts. These results suggest that there is a polarization penalty for online content.

Importantly, this effect was only identifiable through this selected framework as a baseline OLS model could not find a significant result.

A heterogeneity analysis shows that the polarization penalty is not uniform. Namely, there is an asymmetric V-shaped pattern centered around political affiliation and salience. Neutral posts tend to remain relatively insulated from the penalty whereas the penalty harshens as content becomes more political. Moreover, left-wing content experiences a sharper penalty compared with right-wing content.

Further, I identify different penalties based on the tone of posts. Posts with high-urgency or self or group-referencing language experience a larger polarization penalty than their counterparts. This result suggests that users may prefer impartial or objective rhetoric.

Overall, these findings imply a counter-argument to the assumption that social media platforms promote partisan or extreme content. Rather engagement is context specific and depends on the tone of language used. Moreover, they suggest that users tend to prefer informative, balanced content during periods of uncertainty such as the COVID-19 pandemic. Moreover, highly partisan or polarized content does not consistently lead to higher engagement and may actually be penalized by users. Additionally, these results emphasize the importance of moving beyond linear models when using social media data as interactions between variables can be complex and dynamic.

It should be noted that these results are specific to Reddit, the COVID-19 pandemic, and popular subreddits. Thus, these results do not necessarily extend to other platforms, periods, or smaller subreddits. Further, there are a few sources of potential measurement error: some polarizing posts may have been excluded as a result of dropping deleted posts; many posts have only the title and not the body in the dataset; and classification of political affiliation is model-based. However, these types of measurement error are likely to attenuate estimated effects, suggesting these results present a lower bound of the magnitude of the relationship.

Future research should look to assess if these results apply to different social media platforms, smaller subreddits, or different time periods. Additionally, varied definitions of political affiliation or engagement could also be included in future studies.

VI. References

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